

List of Java Programming Lab Programs based on Basic Java Concepts, Control Statements, Java Keywords, Primitive Data Types, Variables, Type Conversion, and Type Casting.

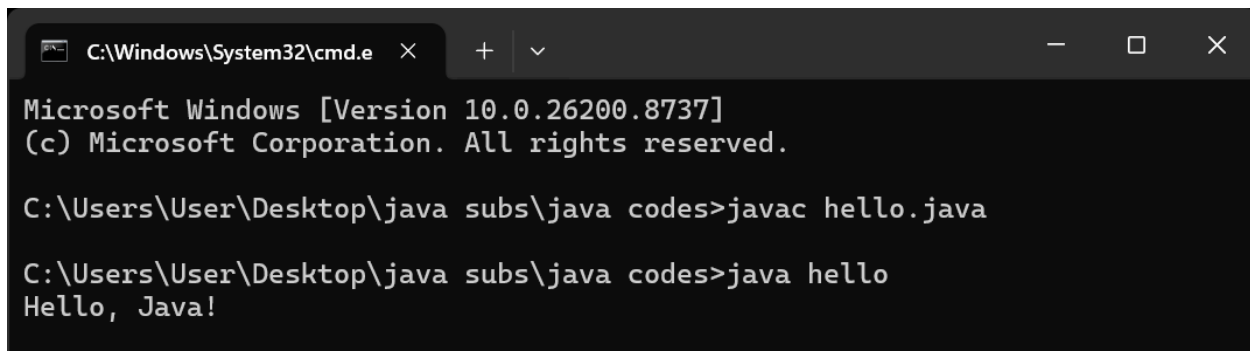
Program 1: Hello Java

Aim:

To write and execute a Java program that displays "Hello, Java!" on the console.

Procedure:

1. Create a Java file named Hello.java.
2. Write the program to print "Hello, Java!".
3. Compile the program using the javac command.
4. Execute it using the java command.
5. Observe the output.



```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User\Desktop\java subs\java codes>javac hello.java

C:\Users\User\Desktop\java subs\java codes>java hello
Hello, Java!
```

Result:

The program was executed successfully and displayed "Hello, Java!" on the screen.

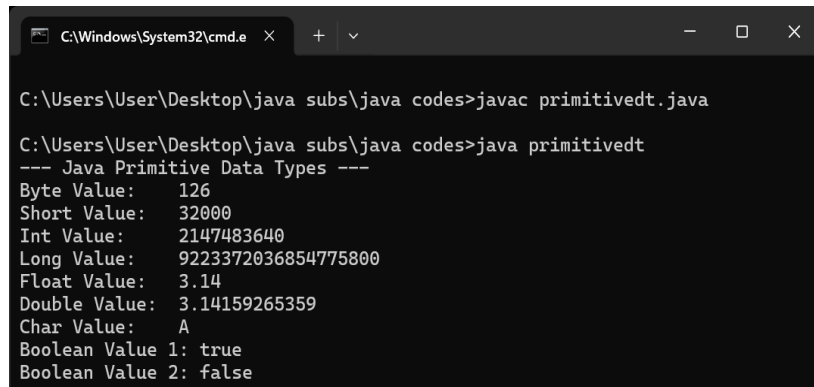
Program 2: Primitive Data Types

Aim:

To declare variables of different primitive data types and display their values.

Procedure:

1. Declare variables of type byte, short, int, long, float, double, char, and boolean.
2. Assign suitable values to each variable.
3. Display the values using output statements.
4. Compile and execute the program.



```
C:\Windows\System32\cmd.e x + v
C:\Users\User\Desktop\java subs\java codes>javac primitivedt.java

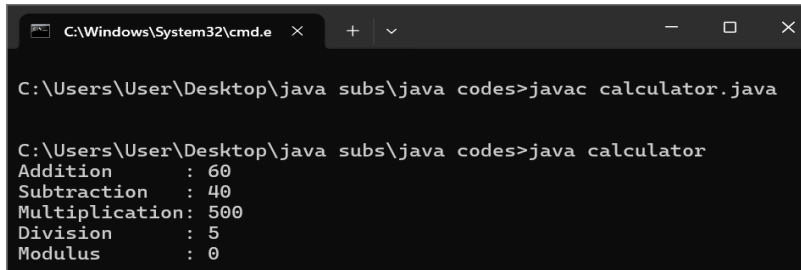
C:\Users\User\Desktop\java subs\java codes>java primitivedt
--- Java Primitive Data Types ---
Byte Value: 126
Short Value: 32000
Int Value: 2147483640
Long Value: 9223372036854775800
Float Value: 3.14
Double Value: 3.14159265359
Char Value: A
Boolean Value 1: true
Boolean Value 2: false
```

Result:

The values of all primitive data types were displayed successfully.

Program 3: Arithmetic Operations on Integers**Aim:**

To perform arithmetic operations on two integer variables.

Procedure:

```
C:\Windows\System32\cmd.e  x  +  v  -  □  ×  
  
C:\Users\User\Desktop\java subs\java codes>javac calculator.java  
  
C:\Users\User\Desktop\java subs\java codes>java calculator  
Addition      : 60  
Subtraction    : 40  
Multiplication : 500  
Division       : 5  
Modulus        : 0
```

1. Declare two integer variables.
2. Perform addition, subtraction, multiplication, division, and modulus operations.
3. Display the results.
4. Compile and run the program.

Result:

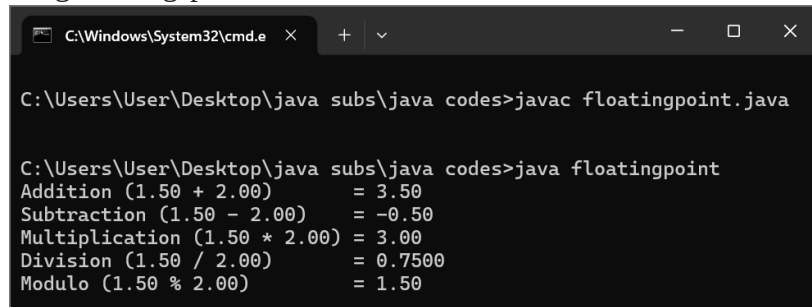
The arithmetic operations on integer variables were performed successfully.

Program 4: Arithmetic Operations on Floating-Point Numbers**Aim:**

To perform arithmetic operations using floating-point numbers.

Procedure:

1. Declare two floating-point variables.
2. Perform basic arithmetic operations.
3. Display the results.
4. Compile and execute the program.



```
C:\Windows\System32\cmd.e  x  +  v  -  □  ×  
  
C:\Users\User\Desktop\java subs\java codes>javac floatingpoint.java  
  
C:\Users\User\Desktop\java subs\java codes>java floatingpoint  
Addition (1.50 + 2.00)    = 3.50  
Subtraction (1.50 - 2.00) = -0.50  
Multiplication (1.50 * 2.00) = 3.00  
Division (1.50 / 2.00)    = 0.7500  
Modulo (1.50 % 2.00)      = 1.50
```

Result:

The arithmetic operations on floating-point numbers were executed successfully.

Program 5: Character and ASCII/Unicode Value**Aim:**

To display a character and its corresponding ASCII/Unicode value.

Procedure:

```
C:\Windows\System32\cmd.e x + v - □ ×

C:\Users\User\Desktop\java subs\java codes>javac chardemo.java

C:\Users\User\Desktop\java subs\java codes>java chardemo
Character: A => ASCII/Unicode Value: 65
Character: B => ASCII/Unicode Value: 66
```

1. Declare a character variable.
2. Convert the character to its ASCII/Unicode value.
3. Display both the character and its value.
4. Compile and run the program.

Result:

The character and its corresponding ASCII/Unicode value were displayed successfully.

Program 6: Boolean Variables

Aim:

To demonstrate the use of Boolean variables and Boolean expressions.

Procedure:

1. Declare Boolean variables.
2. Perform simple Boolean expressions.
3. Display the results.
4. Compile and execute the program.

```
C:\Windows\System32\cmd.e x + v - □ ×

C:\Users\User\Desktop\java subs\java codes>javac booleandemo.java

C:\Users\User\Desktop\java subs\java codes>java booleandemo
x > y: false
x == y: false
x != y: true
```

Result:

The Boolean values and expressions were displayed successfully.

Program 7: Variables

Aim:

To declare, initialize, and display different types of variables.

```
C:\Windows\System32\cmd.e x + v - □ ×

C:\Users\User\Desktop\java subs\java codes>javac variable.java

C:\Users\User\Desktop\java subs\java codes>java variable
Byte Value :125
Short Value :32000
Integer Value (Age) :28
Long Value :9876543210
Float Value (Pi) :3.14
Double Value (Balance) : $54321.85
Char Value (Grade) : A
Boolean Value (Is Java Fun?): true
String Value : Hello, World!
```

Procedure:

1. Declare variables of different data types.
2. Initialize them with suitable values.
3. Display the values.
4. Compile and execute the program.

Result:

The values of different variables were displayed successfully.

Program 8: Swapping Two Variables

Aim:

To swap the values of two variables using a temporary variable.

Procedure:

1. Declare two variables.
2. Use a temporary variable to swap their values.
3. Display the values before and after swapping.
4. Compile and run the program.

Result:

The values of the two variables were swapped successfully.

```
C:\Windows\System32\cmd.e x + v
C:\Users\User\Desktop\java subs\java codes>javac swapping.java
C:\Users\User\Desktop\java subs\java codes>java swapping
Before Swapping:
Value 1 = 5
Value 2 = 10
after Swapping:
Value 1 = 10
Value 2 = 5
```

Program 9: Widening Type Conversion

Aim:

To demonstrate automatic (widening) type conversion.

```
C:\Windows\System32\cmd.e x + v
C:\Users\User\Desktop\java subs\java codes>javac widconver.java
C:\Users\User\Desktop\java subs\java codes>java widconver
Original byte value: 42
Widened to short: 42
Widened to int: 42
Widened to long: 42
Widened to float: 42.0
Widened to double: 42.0
Original char value: A
Widened to int: 65
```

Procedure:

1. Declare variables of smaller data types.
2. Assign them to larger data types.
3. Display the converted values.
4. Compile and execute the program.

Result:

Automatic conversion was performed successfully.

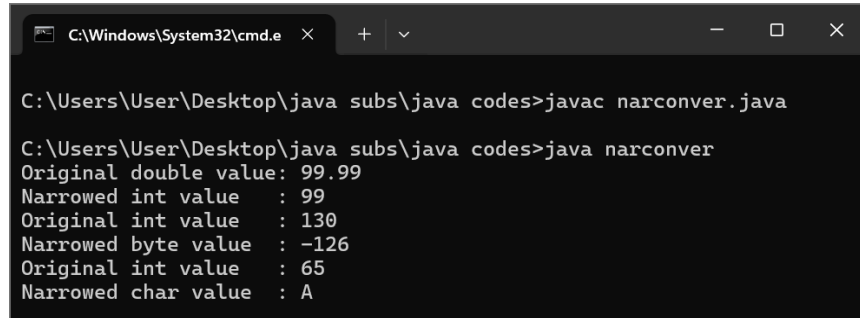
Program 10: Narrowing Type Casting

Aim:

To demonstrate explicit (narrowing) type casting.

Procedure:

1. Declare a variable of a larger data type.
2. Convert it to a smaller data type using type casting.
3. Display the converted value.
4. Compile and run the program.



```
C:\Windows\System32\cmd.e  X  +  v  -  □  X

C:\Users\User\Desktop\java subs\java codes>javac narconver.java

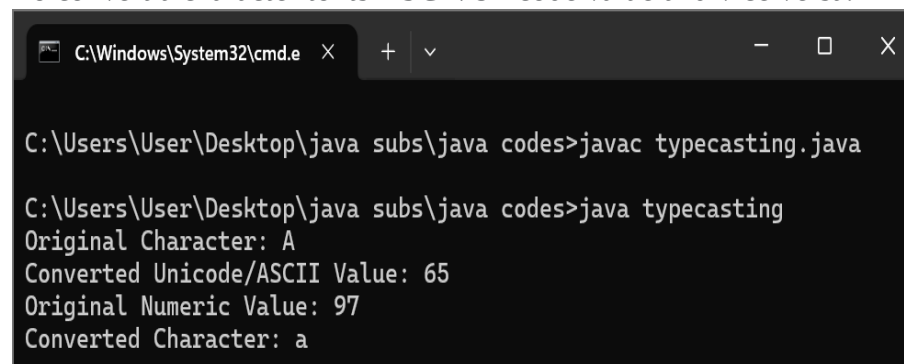
C:\Users\User\Desktop\java subs\java codes>java narconver
Original double value: 99.99
Narrowed int value      : 99
Original int value      : 130
Narrowed byte value     : -126
Original int value      : 65
Narrowed char value     : A
```

Result:

Explicit type casting was demonstrated successfully.

Program 11: Character to ASCII/Unicode Conversion**Aim:**

To convert a character to its ASCII/Unicode value and vice versa.



```
C:\Windows\System32\cmd.e  X  +  v  -  □  X

C:\Users\User\Desktop\java subs\java codes>javac typecasting.java

C:\Users\User\Desktop\java subs\java codes>java typecasting
Original Character: A
Converted Unicode/ASCII Value: 65
Original Numeric Value: 97
Converted Character: a
```

Procedure:

1. Declare a character and an integer value.
2. Perform type casting between character and integer.
3. Display the converted values.
4. Compile and execute the program.

Result:

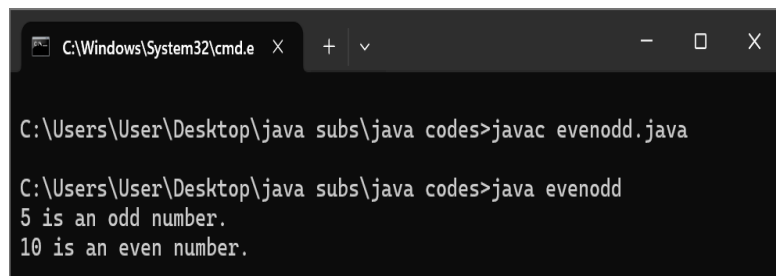
The character and ASCII/Unicode conversions were performed successfully.

Program 12: Even or Odd**Aim:**

To check whether a given number is even or odd using the if-else statement.

Procedure:

1. Declare an integer variable.
2. Use an if-else statement to check whether the number is even or odd.
3. Display the result.
4. Compile and run the program.



```
C:\Windows\System32\cmd.e  X  +  v  -  □  X

C:\Users\User\Desktop\java subs\java codes>javac evenodd.java

C:\Users\User\Desktop\java subs\java codes>java evenodd
5 is an odd number.
10 is an even number.
```

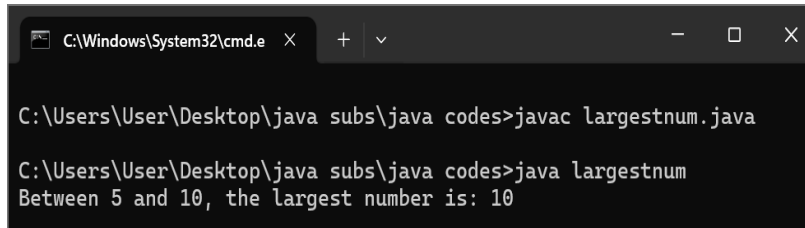
Result:

The program correctly identified whether the number was even or odd.

Program 13: Largest of Two Numbers

Aim:

To find the largest of two numbers using the if-else statement.



```
C:\Windows\System32\cmd.e
C:\Users\User\Desktop\java subs\java codes>javac largestnum.java
C:\Users\User\Desktop\java subs\java codes>java largestnum
Between 5 and 10, the largest number is: 10
```

Procedure:

1. Declare two integer variables.
2. Compare them using if-else statements.
3. Display the larger number.
4. Compile and execute the program.

Result:

The larger of the two numbers was displayed successfully.

Program 14: Java Reserved Keywords

Aim:

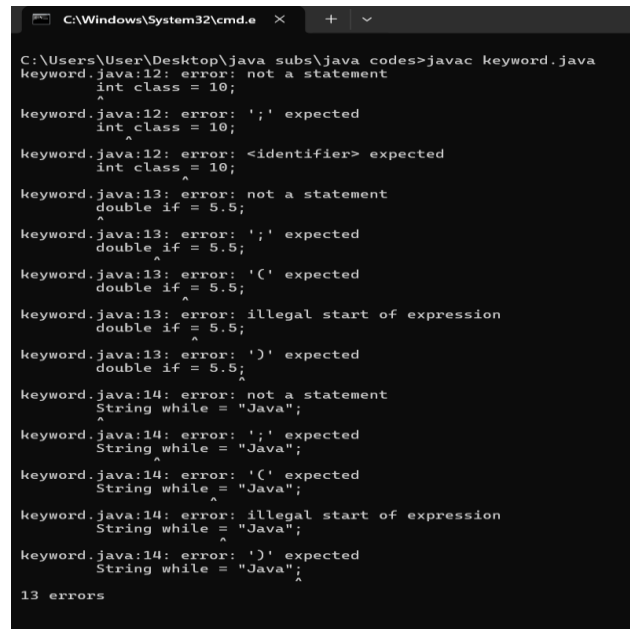
To demonstrate the use of Java reserved keywords and understand why they cannot be used as variable names.

Procedure:

1. Create valid variable names.
2. Observe that Java keywords cannot be used as identifiers.
3. Compile and execute the program.
4. Note the output or compiler behavior.

Result:

The program demonstrated the use of valid identifiers and the restriction on Java keywords.



```
C:\Windows\System32\cmd.e
C:\Users\User\Desktop\java subs\java codes>javac keyword.java
keyword.java:12: error: not a statement
    int class = 10;
    ^
keyword.java:12: error: ';' expected
    int class = 10;
    ^
keyword.java:12: error: <identifier> expected
    int class = 10;
    ^
keyword.java:13: error: not a statement
    double if = 5.5;
    ^
keyword.java:13: error: ';' expected
    double if = 5.5;
    ^
keyword.java:13: error: '(' expected
    double if = 5.5;
    ^
keyword.java:13: error: illegal start of expression
    double if = 5.5;
    ^
keyword.java:13: error: ')' expected
    double if = 5.5;
    ^
keyword.java:14: error: not a statement
    String while = "Java";
    ^
keyword.java:14: error: ';' expected
    String while = "Java";
    ^
keyword.java:14: error: '(' expected
    String while = "Java";
    ^
keyword.java:14: error: illegal start of expression
    String while = "Java";
    ^
keyword.java:14: error: ')' expected
    String while = "Java";
    ^
13 errors
```

Program 15: For Loop

Aim:

To demonstrate the use of the for loop by displaying numbers from 1 to 10.

```
C:\Windows\System32\cmd.e  x  +  v  -  □  x
C:\Users\User\Desktop\java subs\java codes>javac looping.java
C:\Users\User\Desktop\java subs\java codes>java looping
1
2
3
4
5
6
7
8
9
10
```

Procedure:

1. Write a for loop to iterate from 1 to 10.
2. Display each number.
3. Compile and execute the program.
4. Observe the output.

Result:

The numbers from 1 to 10 were displayed successfully.